Data Analyst role

Sunday, 26 July 2026 5:44 pm

Key responsibilities

A Data Analyst may:

- Collect, clean and analyse data.
- Write SQL queries to extract information.
- Create reports and dashboards.
- Identify trends, risks and data-quality problems.
- Support business decisions with accurate information.
- Explain findings to managers and stakeholders.
- Prepare documentation and improve reporting processes.

Required skills

- SQL and relational databases
- Microsoft Excel
- Power BI and Power Query
- Basic DAX and data modelling
- Data cleaning and validation
- Basic statistics
- Analytical and problem-solving skills
- Strong written and verbal communication
- Attention to detail
- Stakeholder consultation
- Time management and prioritisation
- Ability to work with culturally diverse teams

Tools and technologies

Category	**Tools and technologies**	**Purpose**
Data querying	SQL	Extract, filter, join and summarise data stored in databases.
Spreadsheets	Microsoft Excel	Organise, calculate, clean and perform basic analysis on data.
Reporting	Power BI	Create interactive dashboards, charts and business reports.
Data cleaning	Power Query	Import, transform and clean data before analysis or reporting.
Calculations	DAX	Create measures, calculated columns and business calculations in Power BI.
Databases	SQL Server, PostgreSQL, MySQL or Oracle	Store, manage and retrieve structured organisational data.
Programming	Python and pandas	Automate data cleaning, analysis and repetitive reporting tasks.
Cloud data	Microsoft Azure or Microsoft Fabric	Store, process, combine and analyse data using cloud services.
Documentation	Microsoft Word, PowerPoint and Visio	Prepare reports, presentations, procedures and process diagrams.
AI assistance	Microsoft Copilot	Help explain formulas, draft SQL queries, summarise findings and support troubleshooting.

Qualifications

Employers may prefer:

- A degree in IT, Information Management, Data Analytics, Computer Science or a related field.
- Experience with databases, reporting and data analysis.
- Evidence of SQL queries, dashboards or data projects.
- Ability to communicate findings to non-technical people.

Three-Week Data Analyst Learning Plan

The goal is to gain a basic understanding of the main Data Analyst tools. I will study one main topic each day and complete a small practical activity.

Week 1 — Excel, SQL and Databases

Day	Skill	Learning activity	Simple result	Learning resource
Monday	Microsoft Excel	Learn tables, sorting, filtering and basic formulas such as SUM and AVERAGE	Create a small sales spreadsheet	Microsoft Excel Training
Tuesday	Microsoft Excel	Learn IF, XLOOKUP, charts and PivotTables	Create a sales summary and chart	Excel Video Training
Wednesday	SQL	Learn SELECT, WHERE, ORDER BY and DISTINCT	Write five simple SQL queries	Microsoft Learn: Introduction to SQL
Thursday	SQL	Learn GROUP BY, COUNT, SUM and basic joins	Create a summary of sales by location	Microsoft Learn: Query Relational Data
Friday	Relational databases	Learn about tables, rows, columns, primary keys and relationships	Draw a simple customer and order database	Relational Data Concepts

Microsoft Learn provides beginner modules covering relational databases and basic SQL queries.

Week 1 outcome

By the end of Week 1, I should be able to:

- Organise and analyse basic information in Excel.
- Write simple SQL queries.
- Understand how relational databases store information.
- Create a basic chart and PivotTable.

Week 2 — Power Query, Power BI and DAX

Day	Skill	Learning activity	Simple result	Learning resource
Monday	Power Query	Import a CSV file, remove empty rows and change data types	Produce one cleaned dataset	Clean Data with Power Query
Tuesday	Power BI	Learn the Power BI interface and import Excel or CSV information	Create a basic Power BI report	Get Started with Power BI Desktop
Wednesday	Power BI	Create bar charts, cards, tables and slicers	Create a one-page sales dashboard	Beginner Power BI Training
Thursday	DAX	Learn basic measures using SUM, COUNT and AVERAGE	Add three measures to the dashboard	Beginner DAX Workshop
Friday	Data visualisation	Review chart titles, labels, filters and dashboard layout	Improve the dashboard so it is easy to understand	Power BI Training

Microsoft's Power BI beginner resources cover importing data, creating data models, building visualisations and adding basic DAX measures.

Week 2 outcome

By the end of Week 2, I should be able to:

- Clean information with Power Query.
- Import information into Power BI.
- Create a simple dashboard.
- Add basic DAX measures.
- Present information using suitable charts.

Week 3 — Python, Cloud Data, Documentation and AI

Day	Skill	Learning activity	Simple result	Learning resource
Monday	Python	Learn variables, lists, functions and simple calculations	Write a small Python program	Kaggle Python Course
Tuesday	pandas	Import a CSV file, select columns and calculate totals and averages	Create a simple data-analysis notebook	Kaggle pandas Course
Wednesday	Microsoft Fabric	Learn the basic purpose of cloud storage, data pipelines and reporting	Complete one beginner Fabric module	Microsoft Fabric Training
Thursday	Documentation	Write a short report explaining the data, method, findings and limitations	Create a one-page Word report and PowerPoint summary	Microsoft 365 Training
Friday	Microsoft Copilot and final project	Ask Copilot to explain one SQL query and suggest improvements to the report, then check the answers	Complete and review the final mini-project	Microsoft Copilot Beginner Module

project	report, then check the answers	Module

The Kaggle pandas course is available at no cost, while Microsoft provides beginner Fabric and Copilot learning modules.

Week 3 outcome

By the end of Week 3, I should be able to:

- Use basic Python and pandas.
- Understand the purpose of Microsoft Fabric.
- Write a short data-analysis report.
- Use Copilot to support learning and troubleshooting.
- Check AI answers before using them.